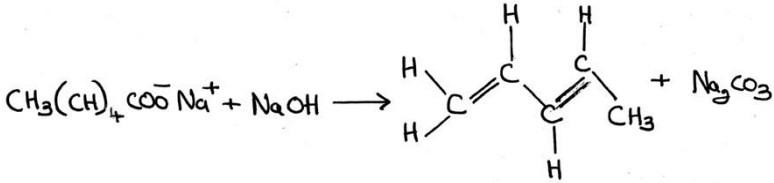
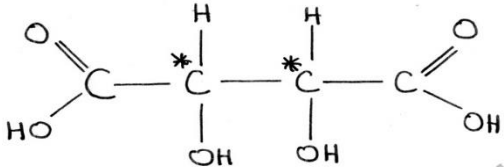


Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
8.	(a)	(i)		200cm ³ at 100°C contains 8.0 g 200cm ³ at 20°C contains 0.32 g (1) amount precipitated = 8.0 – 0.32 = 7.68 / 7.7 g (1) answer must be to two or three significant figures		2		1		2
		(ii)	I	loss of CO ₂ / removal of a carboxyl (or COO) group	1				1	1
			II	award (1) for balanced equation award (1) for structure of the <i>E</i> form of the diene 			2			2
			III	0.4	1					1
	(b)	(i)		 award (1) for either chiral centre identified	1					1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
		(ii)		enantiomers (1) neutral answer – stereoisomers racemic mixture / racemate (1)	2					2
	(c)			(stereoisomerism occurs when two forms of a compound have) the same structure but different positions in space / different 3D arrangement	1					1
	(d)			$\begin{array}{c} \text{HOOC}-\text{C}=\text{C}-\text{COOH} \\ \quad \\ \text{H} \quad \text{H} \end{array} + \text{H}_2\text{O} + [\text{O}] \rightarrow \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{HOOC}-\text{C}-\text{C}-\text{COOH} \\ \quad \\ \text{OH} \quad \text{OH} \end{array}$			1			1
	(e)			polar OH / COOH groups dissolve by hydrogen bonding (1) but as the hydrocarbon chain length increases the polar groups form an increasingly smaller part of a largely non-polar molecule (1)	1	1				2
				Total question 8	7	3	3	1	1	13